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ORIGINAL RESEARCH

Clinical and radiographic outcome of the root canal treatment of infected teeth with associated sinus tract: A retrospective study

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Introduction

Chronic apical abscess is a form of apical periodontitis caused by bacterial infection of the root canal and characterised by a periapical radiolucency associated with an intraoral or extraoral sinus tract (1). Discharge of purulent exudate from the chronic abscess can occur in active sinus tracts, which makes the condition usually asymptomatic. However, the patient may complain of symptoms in the event the sinus tract closes or is blocked.

Histopathologically, lesions associated with sinus tracts can be diagnosed as an abscess or even a granuloma or cyst with focal abscessed areas (2,3). These findings can be possibly explained by the theories about the pathogenesis of the chronic apical abscess, which has been claimed to result from a slight subclinical exacerbation of a chronic lesion (granuloma or cyst) or a chronicisation of an acute abscess (1,3,4).

The prevalence of apical periodontitis lesions associated with sinus tracts has been shown to vary from 8.5% to 18% (5,6). Sinus tracts have been reported to occur more commonly intraorally on the buccal mucosa of teeth with large radiolucencies (2,5). They are usually associated with longstanding root canal infection (7). Teeth with chronic apical abscesses and sinus tracts show a very complex infectious pattern in the apical root canal system, with predominance of biofilms, in many cases extending to the outer root surface to form an extraradicular infection (3).

Many studies have reported that the presence of a sinus tract does not significantly influence the prognosis of the endodontic treatment of teeth with apical periodontitis (6,8–10). However, a large cohort study showed a worse prognosis for teeth with sinus tracts than teeth without a sinus tract (11). In addition to this inconsistency in results, there has been no clear correlation between healing of the sinus tract in a few days or weeks after treatment and the healing of the apical periodontitis lesion one or more years later.

The purpose of this retrospective study was to report on the clinical and radiographic outcome of the root canal treatment or retreatment of teeth with chronic apical abscesses with an intraoral sinus tract performed consecutively by a single experienced operator. The demographic data of these consecutive cases are reported. Further, the influence of some preoperative, intraoperative and postoperative variables on the treatment/retreatment outcome of teeth with sinus tracts was also evaluated.

Material and methods

Case selection

Cases included in this study were selected from 144 patients (81 women and 63 men; age ranging from 15 to 86 years) who were referred to a specialist in Endodontics (L.A.), in a private practice, for endodontic evaluation because of the presence of an intraoral sinus tract. All these referred patients showed up consecutively over the period from 1999 to 2019. Initially, case selection followed inclusion criteria as follows: tooth with a sinus tract and radiographic evidence of periradicular bone destruction, tooth with a necrotic pulp confirmed by pulp tests or with previous root canal treatment, tooth that was restorable and had an indication for root canal treatment or retreatment, tooth with a mobility score <3 and absence of periodontal pockets >4 mm deep, and patients without significant systemic disease. All patients were informed about the treatment options, as well as the benefits and risks associated with each one, and gave consent for examination and treatment of their teeth. The approval of this retrospective study protocol was obtained from the Institutional Board Review.

Data collection

The following preoperative clinical and radiographic data were recorded: patient's age and sex, sinus tract location (buccal, distal, palatal/lingual), size of the radiolucent lesion, root canal status (necrotic pulp or previous treatment), tooth type and location (maxillary or mandibular), symptoms, periodontal probing and tooth mobility. The size of the apical periodontitis lesion was measured on periapical radiographs as the largest diameter and was classified as small (≤5 mm), large (from >5 to <10 mm) or very large (≥10 mm).

Intraoperative data were also recorded, including the number of treatment sessions, working length, obturation technique, quality of the root canal filling and filling of lateral canals. The root canal filling was regarded as adequate when all canals were obturated, with no voids in the filling mass and the apical terminus of the filling was 0–1 mm short of the radiographic apex.

Endodontic treatment/retreatment procedures

Of the cases with preoperative presence of a sinus tract, 137 teeth from 128 patients met the inclusion criteria and were subjected to root canal treatment or retreatment by a single operator (L.A.). In all cases, thermal pulp tests were used to evaluate the pulp diagnosis and a gutta–percha point was used to follow the pathway of the sinus tract down to the involved tooth and radiographed. Periodontal probing, tooth mobility assessment and evaluation of fissures and fractures were also performed. In the first treatment session, local anaesthesia was administered and the tooth was isolated with a rubber dam. After disinfection of the operative field, chemomechanical preparation was performed using 2.5% sodium hypochlorite as the irrigant and diverse manual or rotary NiTi instruments, which differed over the years. The working length (WL) was established by using an electronic apex locator, and then radiographically confirmed. Most cases were treated in a single visit, provided that a dry canal was achieved after preparation. An additional visit with interappointment calcium hydroxide medication was made necessary in more anatomically complex cases or when a dry root canal could not be achieved before obturation. Calcium hydroxide had to be renewed for one more session in a few cases when the canal was still wet. Root canals were filled by the lateral compaction or the Schilder's vertical compaction technique.

In retreatment cases, the previous filling material was removed by using Gates–Glidden burs and Hedström files. Gutta–percha solvent (xylol, Dickinson, Buenos Aires, Argentina) was used in most cases. Afterwards, the root canal procedures were conducted as described for the initial treatment.

Follow-up

Healing of the sinus tract was recorded during follow-up examinations performed at 7, 14, 21 and 30 days postoperatively. The result was dichotomous, that is healed (closed) or not healed (active sinus tract). Persistent cases were recorded as those that did not heal after 30 days. Recurrent cases were those in that the sinus tract re-emerged any time after having healed.

Regarding periradicular tissue healing, patients were followed up periodically (range, from 1 to 17 years). Appearance or persistence of any clinical signs or symptoms of apical periodontitis were indicative of a failed outcome, even if they occurred in less than one-year follow-up. Patients were contacted by telephone every year after treatment for recall examination. At the follow-up visit, patients were evaluated for the presence of clinical signs and symptoms, such as sinus tract, spontaneous pain, tenderness to percussion/palpation, root fracture, loss of function, mobility and periodontal probing depths. The presence, type and quality of the coronal restoration at the time of follow-up examination was evaluated by visual inspection and radiographs and included as a postoperative variable. Coronal restoration was regarded as adequate when it appeared intact, showing no detectable defects or recurrent caries.

For radiographic follow-up examination, preoperative, postoperative and follow-up radiographs were compared to evaluate the fate of the apical periodontitis lesion. For evaluation, radiographs were incorporated in Keynote files and examined on the computer screen, in a darkened room, independently by 2 blinded evaluators, who are experienced endodontists. The treatment provider did not participate of the evaluations. The periradicular status was evaluated on the basis of the Periapical Index (PAI) system (12). Joint discussion was used to score the disagreement cases. Observers were calibrated against a set of 100 reference radiographs (kindly supplied by Dag Ørstavik, University of Oslo). The outcome in multirooted teeth was categorised according to the root with the highest PAI score.

According to the clinical and radiographic evaluations, the outcome was classified as:

1 Healed: the tooth was asymptomatic and functional, the sinus tract disappeared, and there was no radiographic apical periodontitis lesion (PAI scores 1 or 2)

2 Healing: the tooth was asymptomatic and functional, the sinus tract disappeared, and the apical periodontitis lesion decreased in size (PAI score decreased, but it was still >2), in follow-up evaluations shorter than 4 years

3 Diseased: the tooth was symptomatic and nonfunctional, the sinus tract persisted or recurred, and/or the radiographic apical periodontitis lesion was unchanged or enlarged (PAI scores 3 to 5).

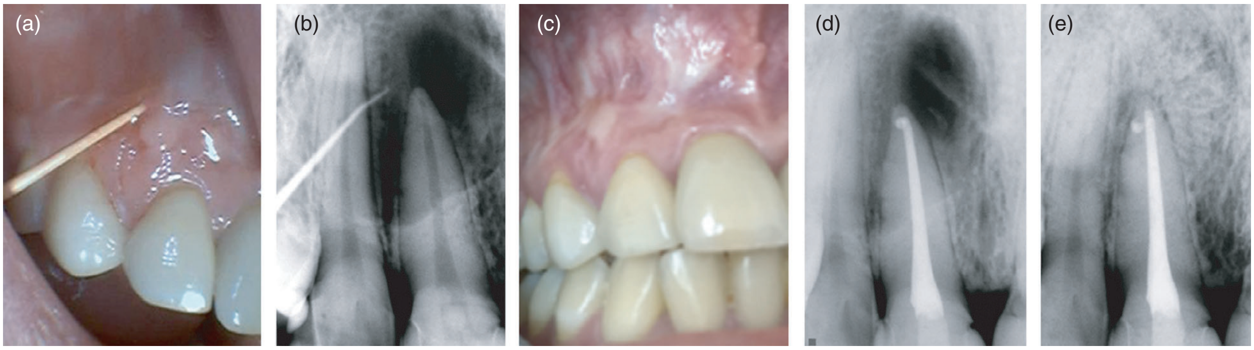
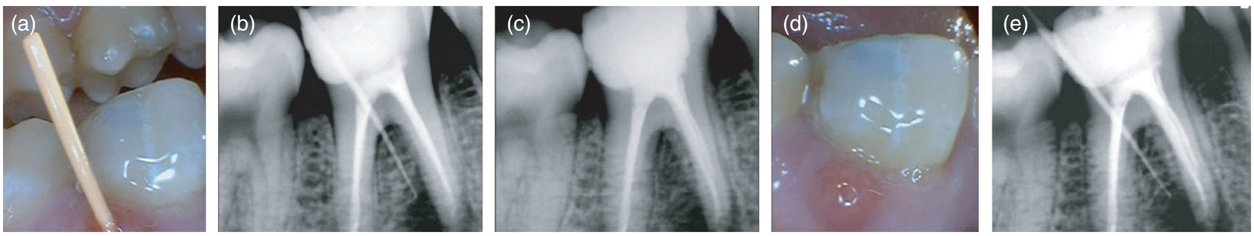


Figure 1 Representative case of a tooth with associated sinus tract categorised as healed. (a and b) Tracing the sinus tract with a gutta–percha point. (c) Postobturation radiograph. (d) Follow-up radiograph showing periradicular bone healing after 2 years.



Conclusion

Findings from the present study indicate that the quick closure of the sinus tract and the later healing of the associated periradicular lesion are both reliable upon effective control of the intraradicular infection by treatment or retreatment. Persistence of a draining sinus tract for more than a month or its recurrence after treatment indicates the presence of a persistent intraradicular/extraradicular infection that needs to be managed by retreatment, surgery or extraction. Closure of the sinus tract in the first month following treatment may be a good predictor of favourable long-term outcome.

Conflict of interest

The authors declare that they have no conflict of interest.

Authorship declaration

All authors have contributed significantly and are in agreement with the manuscript. L.A., designed the research study, and performed the selection and treatment of all cases. A.F.C., and G.S., involved in data tabulation, analysis and revision. F.R.F.A., involved in data and text revision. I.N.R., involved in data revision and wrote the paper. J. F. S. designed the research study, analysed the data and wrote the paper.

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